SHD training smoke test — it trains

exp060 · 11 July 2026 · Draft

Abstract

Before the program plan's baseline can be trusted, the Spiking Heidelberg Digits (SHD) training pipeline has to clear one low bar: does it actually **train**? SHD is a benchmark of spoken digits recorded as cochlea-model spike trains. This run exercises Rung A of the plan, the free signed-recurrent ceiling: the least-constrained model in the ladder, expected to set the performance upper bound, with all four recurrent weight blocks trainable, no Dale's-law sign constraint (weights may take either sign), and the gamma-band rhythm left untuned. It trains that model on a 1000-sample subset for 30 epochs and asks only whether the loss falls and the test accuracy climbs off the 5% chance floor (one correct label in 20 classes, guessed at random). Both bars clear: training loss falls from 3.17 to 0.56, and test accuracy reaches 62%, an order of magnitude above chance. This is engineering validation, *not* a registered result: the honest ceiling number has to come from a full-scale run, where the test set is not swamped by so small a training subset.

Methods

The run, step by step:

1. Load the SHD training set and take a 1000-sample subset of the 8156-utterance training pool.
2. Bin each utterance's input spikes onto the $\Delta t = 1$ ms grid over the $T = 1000$ ms window ($T_{\text{steps}} = 1000$).
3. Build the COBANet model with 256 hidden units, all four recurrent weight blocks trainable and signed (no Dale's law).
4. Train for 30 epochs by backpropagation through time with a surrogate gradient for the non-differentiable spike, under an upper firing-rate regulariser that keeps the signed recurrence from running away.
5. After each epoch, evaluate cross-entropy loss and classification accuracy on the held-out test set, and record the best accuracy.

Parameter	Value
Dataset	SHD, 1000-sample subset (of the 8156-utterance training set)
Model	COBANet (a conductance-based spiking network), 256 hidden units, all four recurrent weight blocks trainable, signed (no Dale's law)
Integration	$\Delta t = 1$ ms, $T = 1000$ ms ($T_{\text{steps}} = 1000$): a coarse timestep, chosen so a full 1 s window still trains in minutes

Training	30 epochs, learning rate 0.001, batch size 32, trained by backpropagation through time (BPTT) with a surrogate gradient for the non-differentiable spike
Regulariser	upper firing-rate bound (threshold $\theta_u = 100$, strength $s_u = 0.06$, values from Cramer et al. [**1**]): without it the signed recurrence runs away

The coarse $\Delta t = 1$ ms is the fastest our conductance model tolerates (verified on MNIST); it is still ≈ 10× finer than the discrete leaky integrate-and-fire (LIF) timestep used by Cramer et al. [**1**], because our fast synaptic and inhibitory time constants ($\tau_{\text{AMPA}} = 2$ ms, $\tau_{m,I} = 5$ ms) need resolving. Where:

- Δt, the integration timestep (the interval each simulation step advances);
- T, the duration of one input utterance in simulated time;
- $T_{\text{steps}} = T/\Delta t$, the number of timesteps the network is unrolled over and backpropagated through per utterance;
- τ_{AMPA}, the excitatory (AMPA) synaptic decay time constant;
- $\tau_{m,I}$, the membrane time constant of the inhibitory population;
- θ_u, the firing rate above which the upper-bound regulariser penalises a neuron;
- s_u, the strength (weight) of that regulariser penalty.

Results

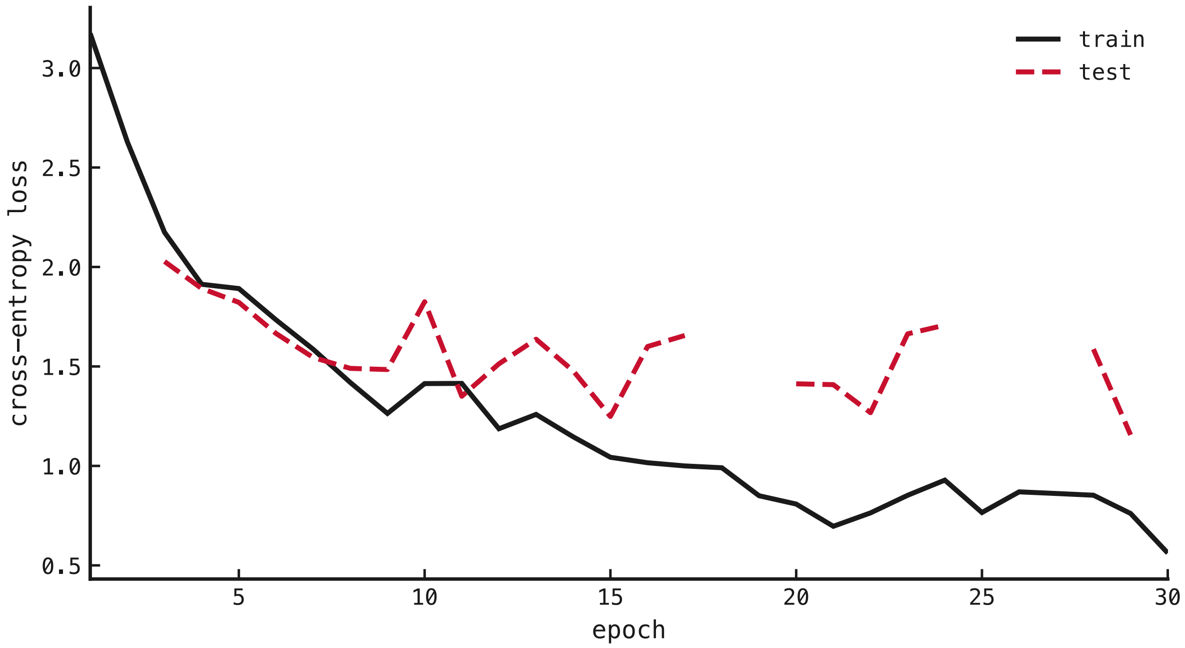

Figure 1: Cross-entropy loss against training epoch, for the training set (solid black) and the held-out test set (dashed red). The horizontal axis is the training epoch; the vertical axis is mean cross-entropy loss. Training loss falls from 3.17 to 0.56 test loss falls with it for the first few epochs, then turns upward as the network overfits the 1000-sample subset.

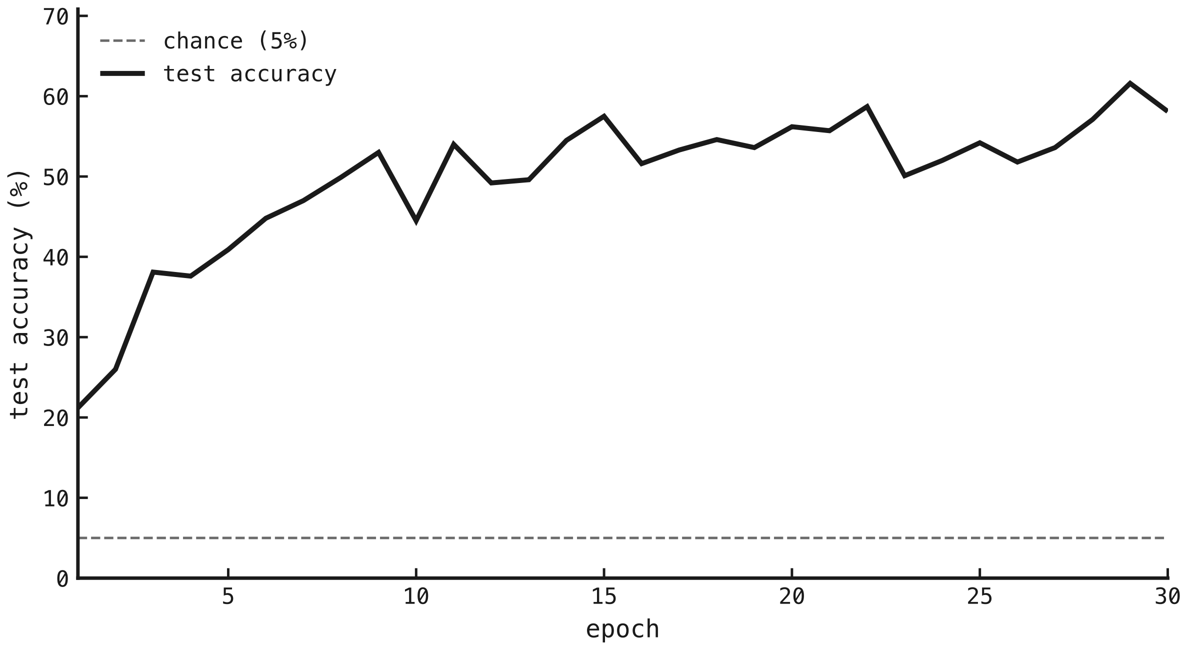

Figure 2: Test accuracy against training epoch. The horizontal axis is the training epoch; the vertical axis is the percentage of held-out test utterances classified correctly. The dashed grey line marks the 5% chance floor (one of 20 classes). Accuracy climbs from 21% at epoch 1 to a best of 62% at epoch 29, an order of magnitude above chance, then plateaus and wobbles (final 58%) as the subset overfits.

Both bars are cleared: accuracy reaches 62%, far above both the 5% floor and the plan's "decisively above chance" threshold of 25%. So the SHD data path, the spike-to-input binning, the all-four-blocks-trainable signed recurrence, and the surrogate-gradient training loop all work end to end. The firing-rate regulariser is load-bearing: without it the earlier unregularised run blew up to a runaway inhibitory firing rate; with it, rates stay physiological and the network learns.

The overfitting is the expected cost of the small subset and is *why* this is a smoke test, not a result: the honest ceiling number has to come from a full-scale run, where the test set is not swamped by a 1000-sample training set. That full-scale run is the plan's registered Rung A. Next: scale the subset up (and add mild early-stopping) for the number, then descend the ladder (the plan's sequence of successively more constrained models) to the Dale's-law-constrained baseline.

References

1. Cramer, Stradmann, Schemmel & Zenke — *The Heidelberg Spiking Data Sets for the Systematic Evaluation of Spiking Neural Networks*. IEEE Transactions on Neural Networks and Learning Systems, 2020. doi:10.1109/TNNLS.2020.3044364